

Minutes of Meeting 2

05/08/2026

Present: Christo, Dylan, Lithasa, Dhava, Murphy

Apologies: Terry

Absentees:

Meeting Start: 1.15pm

Meeting End: 2pm

Time:	Item #	Topic:	Details:	Action:
13:15	1	Project structure	Talked through structure of the project and the chosen repositories to fulfill each role: - Mujoco for simulation - Development on Ubuntu 22 - Camera pose and gesture detection using MediaPipe - G1 Robot current state and interaction using Unitree SDK Python - Kinematics using Pinocchio - For motion capture suit - GMR	Install following repositories on Ubuntu development environment: - Mujoco https://github.com/unitreerobotics/unitree_mujoco - MediaPipe https://github.com/google-ai-edge/mediapipe - Unitree SDK Python https://github.com/unitreerobotics/unitree_sdk2_python - Pinocchio https://github.com/stack-of-tasks/pinocchio - GMR https://github.com/YanjieZe/GMR -
13:25	2	Scope	- Robot's 'read/react rate' will be set to slow. We can assume that the robot's reaction rate should be slow. (i.e. the robot will only react to something once every couple seconds or such; prevents constant stream of instructions opposing instructions) - Project will be delivered as a Github Repository incorporating other repositories as dependencies. The README will include instructions for setting up the environment and running it.	
13:30	3	Ubuntu	Discussed Ubuntu incompatibilities. ARM64 vs amd64 architecture causing different versions of Ubuntu. Agreed that those that need to, should develop on an Ubuntu Virtual Machine. Testing can be done on Mujoco inside VM and then on a clean Ubuntu install (Christo has one on an external drive)	Everyone that still needs to should install Ubuntu on a virtual machine. Programs to install on the VM should include the above dependencies/repositories as well as: - Github - VSCode - Python
13:35	4	Project Acceptance Tests &	By the end of this project: - The robot should be able to identify it's leader and follow them at a constant distance;	Murphy to write Sprint 1 Deliverables Scope Dylan to write Sprint 1 Deliverables Project Acceptance Tests

		Requirement s	- Should recognize it's leader's gestures and react to them (turn around, backward, forward, turn left, turn right, go left, go right); - Should be able to 'mirror' it's leaders actions Requirements to be extracted from scope. Skills and resources also important. Break down broad goals into individual requirements	
13:45	5	Tasks for next 2 weeks	The group of 6 should split into two subgroups. - Group One: to focus on person identification task. Develop a program using MediaPipe that: - o Gets a video as input (can have multiple people) - o Identifies people in the video - o As output: returns a re-rendered video showing a bounding box around the identified people - o Return a skeleton? - Group Two: Gesture recognition task. Develop program using MediaPipe that can translate and output to console the meaning of certain gestures. - o 7 different gestures - o Outputs to console in format: ▪ Frame 56 : Thumbs up	
14:00	6	Meeting end		